Unit: 6 Unit 8: Quadratics
Topic: Systems of Quadratic Equations

Name: ________________
Date: ___________

Advanced (Habanero)

Q1. Solve the system: $x^2 + y^2 = 4$, $x^2 - y^2 = -1$
(A) $(-1, -sqrt3)$
(B) $(1, sqrt3)$
(C) $(-1, sqrt3)$
(D) $(1, -sqrt3)$
(E) No solution

Q2. The system $y = x^2 - 2$, $y = -x^2 + 2$ has solutions at $x =$
(A) $-1, 1$
(B) $-2, 2$
(C) 0
(D) $1, -1$
(E) $2, -2$

Q3. The graphs of $y = x^2 + 1$, $y = -x^2 - 1$ intersect at
(A) $(0, 1)$
(B) $(1, 0)$
(C) $(0, 0)$
(D) $(0, -1)$
(E) No intersection

Q4. The system $y = x^2 - 4$, $y = x^2 - 1$ has solutions at $y =$
(A) $-4, -1$
(B) $-1, 1$
(C) $-3, 3$
(D) $1, -1$
(E) $-2, 2$

Q5. The equation $x^2 + y^2 = 4$ represents a
(A) Circle centered at $(0, 0)$ with radius 2
(B) Circle centered at $(1, 1)$ with radius 2
(C) Ellipse centered at $(0, 0)$
(D) Parabola centered at $(0, 0)$
(E) Hyperbola centered at $(0, 0)$

Q6. Solve the system: $y = x^2 - 1$, $x = -2$
(A) $(-2, 3)$
(B) $(-2, -1)$
(C) $(-2, 1)$
(D) $(-2, -3)$
(E) $(-2, 0)$

Q7. The system $x^2 + y^2 = 1$, $y = x^2 - 1$ has solutions at $x =$
(A) $-1, 1$
(B) $0, 0$
(C) 0
(D) -1
(E) 1

Q8. The system $y = x^2 + 1$, $y = -x^2 - 1$ has
(A) One solution
(B) Two solutions
(C) No solution
(D) Infinitely many solutions
(E) Cannot be determined

Open Ended Questions (FRQ)

Q9. Solve the system $y = x^2 + 2$, $y = x^2 - 2$. Write your answer as an ordered pair (x, y). Use $\boxed{...}$ to enclose your final answer.

Q10. Solve the system $x^2 + y^2 = 9$, $y = x^2 - 4$. Write your answer as an ordered pair (x, y). Use $\boxed{...}$ to enclose your final answer.

Chef's Tips

- Tip 1: Make sure to provide clear and concise answers.
- Tip 2: Use proper notation when writing equations.
- Tip 3: Encourage students to check their work.